

Learning Using Privileged Information for Litter Detection

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Introduction

(Image Source: [1])

A large, sprawling pile of garbage, primarily plastic waste, under a cloudy sky. The waste is a mix of white, clear, and colored plastics, some crumpled and some in larger pieces. The pile extends across the entire width of the image, with a dark green banner at the bottom.

*We often say we want to leave the Earth a better place than we found it...
Yet at this rate, future generations will inherit a planet buried in waste.*

A Step Towards a Cleaner Future

A research-driven approach to sustainability through AI.

Our approach combines *Learning Using Privileged Information (LUPI)* with deep learning *Object Detection (OD)* frameworks to enhance litter detection accuracy.

Improves detection of marginally small and hard-to-detect litter while maintaining model efficiency and scalability.

Problem Definition

DJI Air 2S (Source: [2])



Litter pollution is rising globally, posing environmental and socio-economic challenges.

Detecting small, transparent, or partially hidden litter is difficult with current AI and UAV-based methods.

Existing approaches rely on complex architectures or large datasets, limiting efficiency and scalability.

Our approach: Use *privileged information* to enhance *detection accuracy* without increasing *model complexity* or *inference time*.

Litter Detection

Existing Datasets:

1. **UAV-Bottle / BDW (2018)**: 25,407 **UAV** images, bottle detection.
2. **TACO (2020)**: 1,500 images, 60 categories, broad litter detection.
3. **MJU-WASTE (2020) & UAVWaste (2021)**: Focus on small or single-category objects.
4. **ZeroWaste & PlastOPol**: Real-world datasets, 2–4k images.
5. **SODA (2024)**: 829 **UAV** images, multiple altitudes and categories.

Detection Approaches: *YOLO, Faster R-CNN, SSD, RetinaNet.*

Pre-processing Steps: *Tiling, Augmentation.*

Trend: Improvements in detection accuracy often require **complex models and largely annotated datasets**, highlighting the need for efficient solutions.

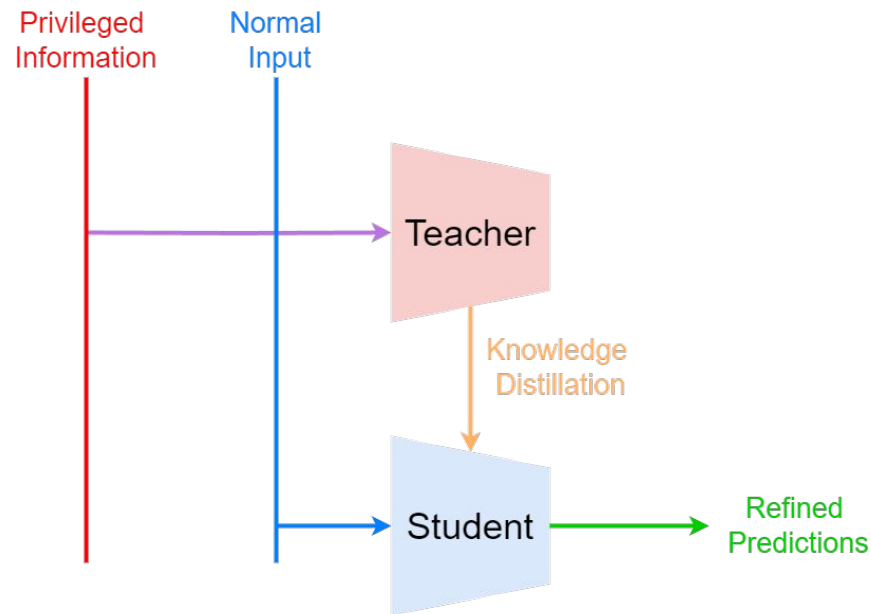
Learning Using Privileged Information

Incorporates **supplementary data** during training to improve prediction accuracy.

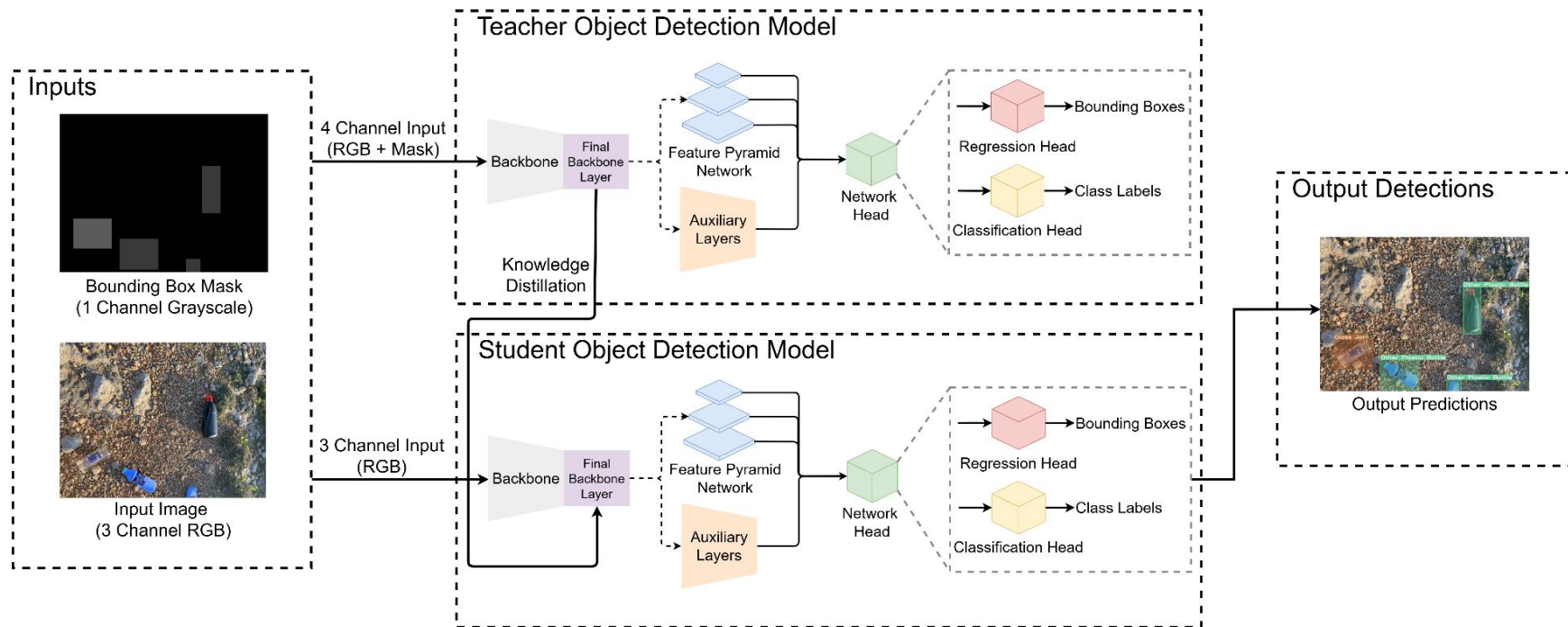
Enables **knowledge transfer** from a teacher with privileged data to a student with standard inputs.

Particularly effective in computer vision tasks with **asymmetric** training and test information.

Improves performance **without increasing model complexity** at inference time.



Proposed Solution

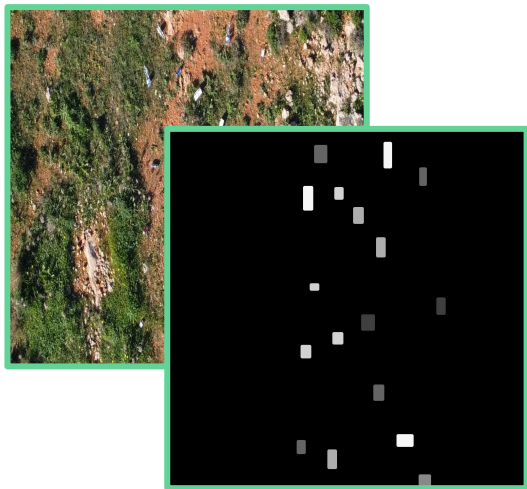


Proposed architecture integrating LUPI into modern deep learning object detection frameworks for litter detection.

Methodology

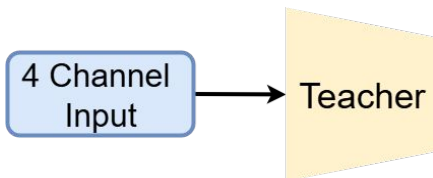
Generate Privileged Information

Generate Bounding Box Mask



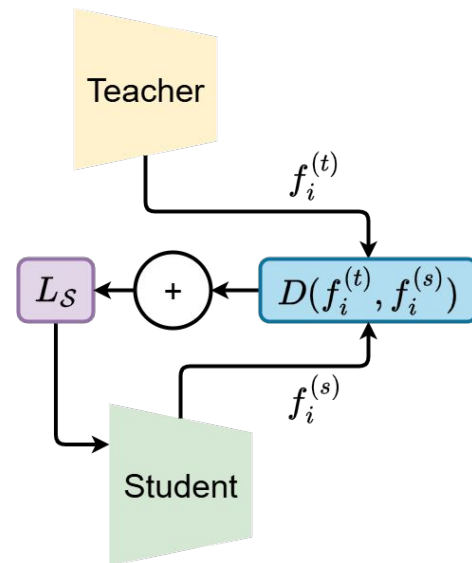
Train Teacher Model

Modify Teacher Input



Distill Knowledge to Student Model

Modify Student Loss Function



Experimental Setup

Datasets:

1. SODA (training, 829 images, varied UAV altitudes);
2. BDW & UAVVaste (cross-dataset evaluation)

Pre-processing: SODA images tiled 3×3, resized to 1280×1280, Min-Max normalization; BDW & UAVVaste resized only

Training & Evaluation: All detectors trained on SODA, evaluated within SODA and across BDW & UAVVaste

Post-processing: Non-Maximum Suppression (IoU=0.5)

Within-Dataset Evaluation

Purpose

- Test LUPI framework for UAV litter detection
- Handle small objects & varied altitudes

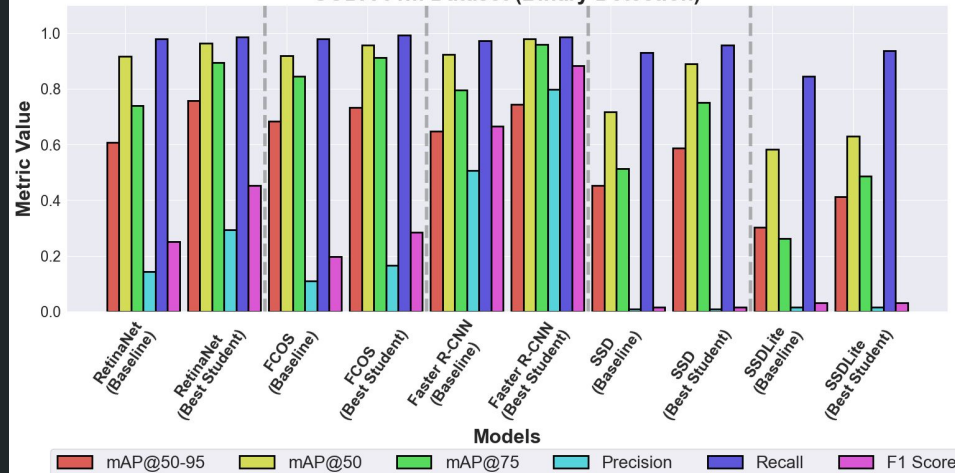
Method

- **3** experiments:
 - Binary at 1 metre (no tiling)
 - Binary across altitudes (3×3 tiling)
 - Multi-label across altitudes (3×3 tiling)
- **5** architectures × teacher + students
- **α** values {0, .25, .5, .75, 1}

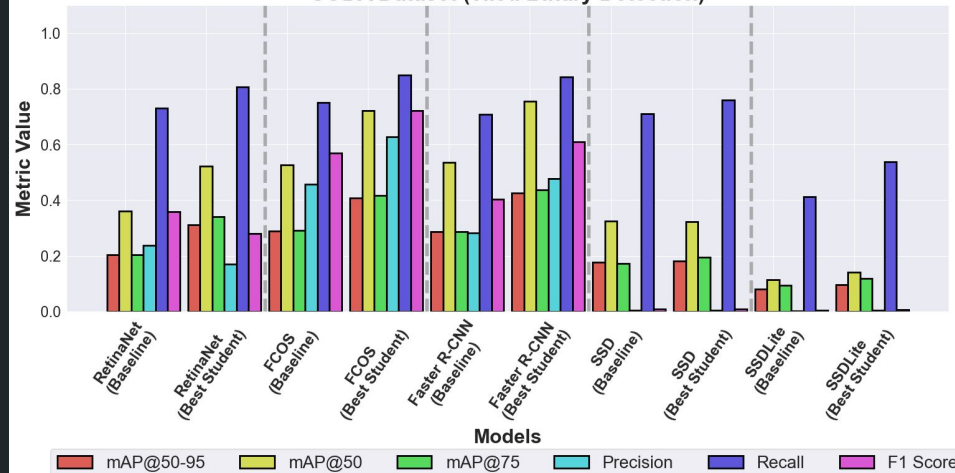
Key Takeaways

- Student model accuracy usually better than the baseline model
- **Best models:** RetinaNet (close-range), Faster R-CNN (multi-alt), FCOS (multi-label)
- Performance boosts come with $\uparrow$ training time, but no extra cost at inference

Comparison of Baseline and Best Student Models Across Key Detection Metrics on SODA 01m Dataset (Binary Detection)



Comparison of Baseline and Best Student Models Across Key Detection Metrics on SODA Dataset (Tiled Binary Detection)



Cross-Dataset Evaluation

Purpose

- Assess generalisation of SODA-trained models
- Compare LUPI-trained students against baselines

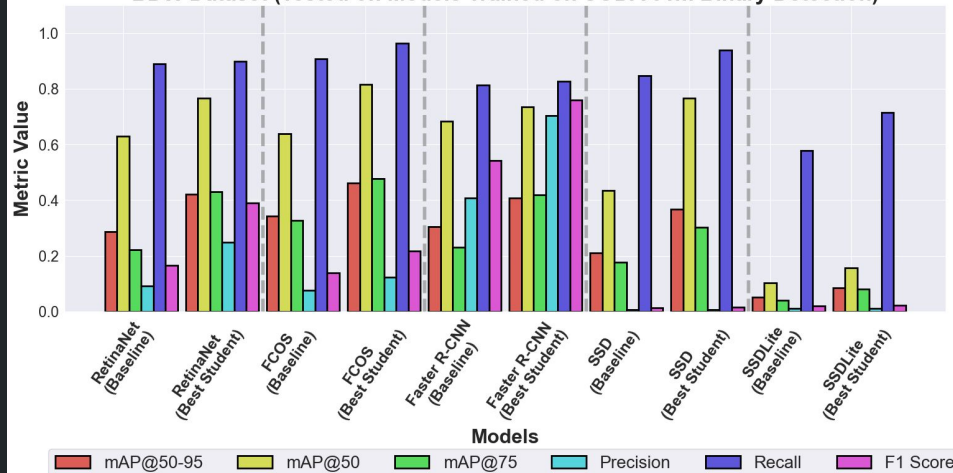
Method

- **BDW**: models from SODA at 1 metre
- **UAVVaste**: models from SODA 3×3 tiling
- Binary litter detection across both datasets

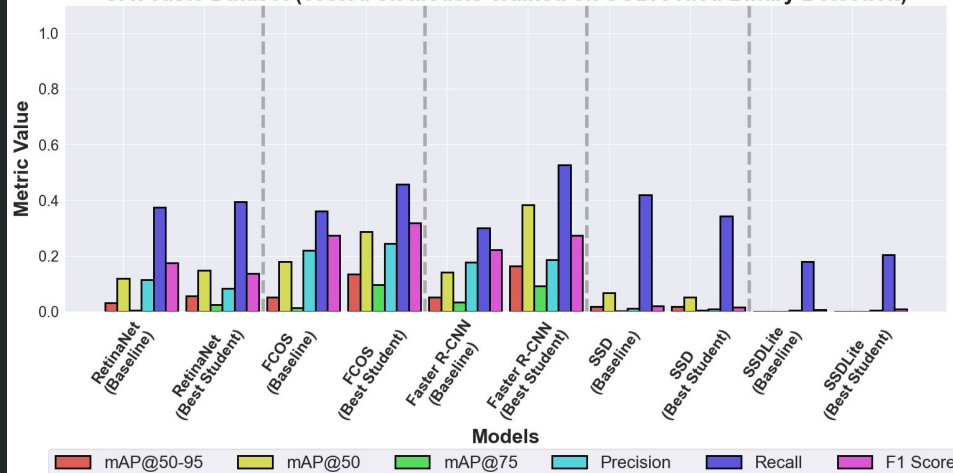
Key Takeaways

- LUPI-trained students consistently outperform baselines
- Strongest results from FCOS, Faster R-CNN, and RetinaNet
- SSD/SSDLite perform poorly, limited transfer
- LUPI improves adaptability without increasing model size

Comparison of Baseline and Best Student Models Across Key Detection Metrics on BDW Dataset (Tested on Models Trained on SODA 01m Binary Detection)



Comparison of Baseline and Best Student Models Across Key Detection Metrics on UAVVaste Dataset (Tested on Models Trained on SODA Tiled Binary Detection)

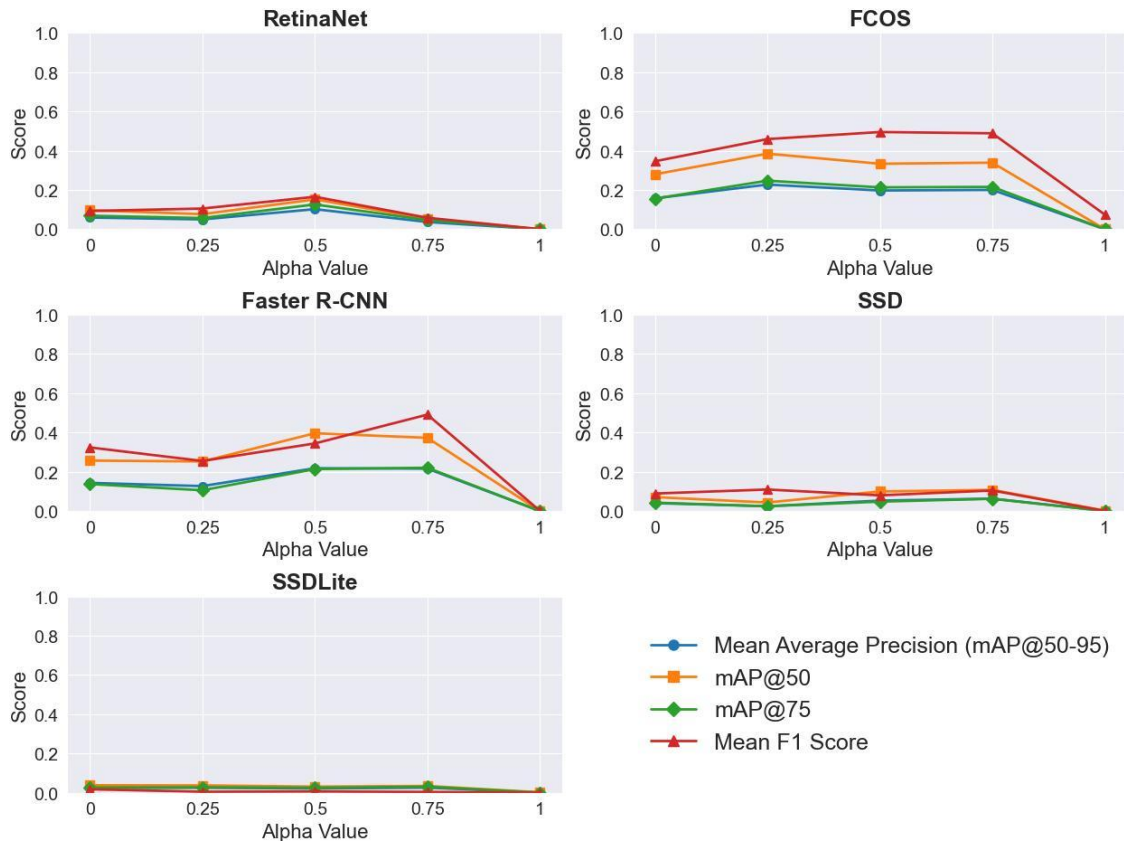


Teacher Model Results

Model	mAP@50-95	mAP@50	mAP@75	Precision	Recall	F1 Score
RetinaNet	0.88	0.92	0.91	0.76	0.97	0.85
FCOS	0.91	0.95	0.94	0.91	0.97	0.94
Faster R-CNN	0.95	0.99	0.98	0.96	0.99	0.97
SSD	0.36	0.49	0.45	0.59	0.76	0.63
SSDLite	0.11	0.13	0.13	0.00	0.37	0.01

Teacher Model Performance on SODA Dataset (Tiled Multi-Label Detection)

Ablation Study on α Parameter



Conclusion

Proposed a methodology *integrating privileged information and knowledge distillation* for litter detection.

Demonstrated improved detection of *small and multi-scale objects* across five renowned object detectors.

Achieved *consistent improvements in accuracy and cross-dataset generalisation* on SODA, BDW, and UAVVaste.

Improvements obtained *without increasing model complexity*, ensuring practical applicability.

Acknowledgements

This work is part of the “**Aerial Waste Identification and Geolocation System (AWIGS)**” project.

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The project aims to develop an **autonomous waste detection and geolocation system** to support a cleaner, more sustainable world.



Academic Team

Aerial Waste identification and Geolocation System



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All You Need Is Already Online

All code and materials are open
source on **GitHub**:



[github.com/mbar0075/
lupi-for-object-detection](https://github.com/mbar0075/lupi-for-object-detection)



Questions and Comments?

They are welcome. . .

References

- [1] NSO Malta, “Municipal Waste: 2022,” 2023. [Online]. Available: <https://nso.gov.mt/municipal-waste-2022-2>. [Accessed: 30-Jan-2025].
- [2] DJI, “DJI Air 2S,” DJI. [Online]. Available: <https://www.dji.com/mt/support/product/air-2s>. [Accessed: 30-Jan-2025].
- [3] D. Pisani, D. Seychell, C. J. Debono and M. Schembri, “SODA: A Dataset for Small Object Detection in UAV Captured Imagery,” *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, United Arab Emirates, pp. 151-157, 2024.



Thank you!

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